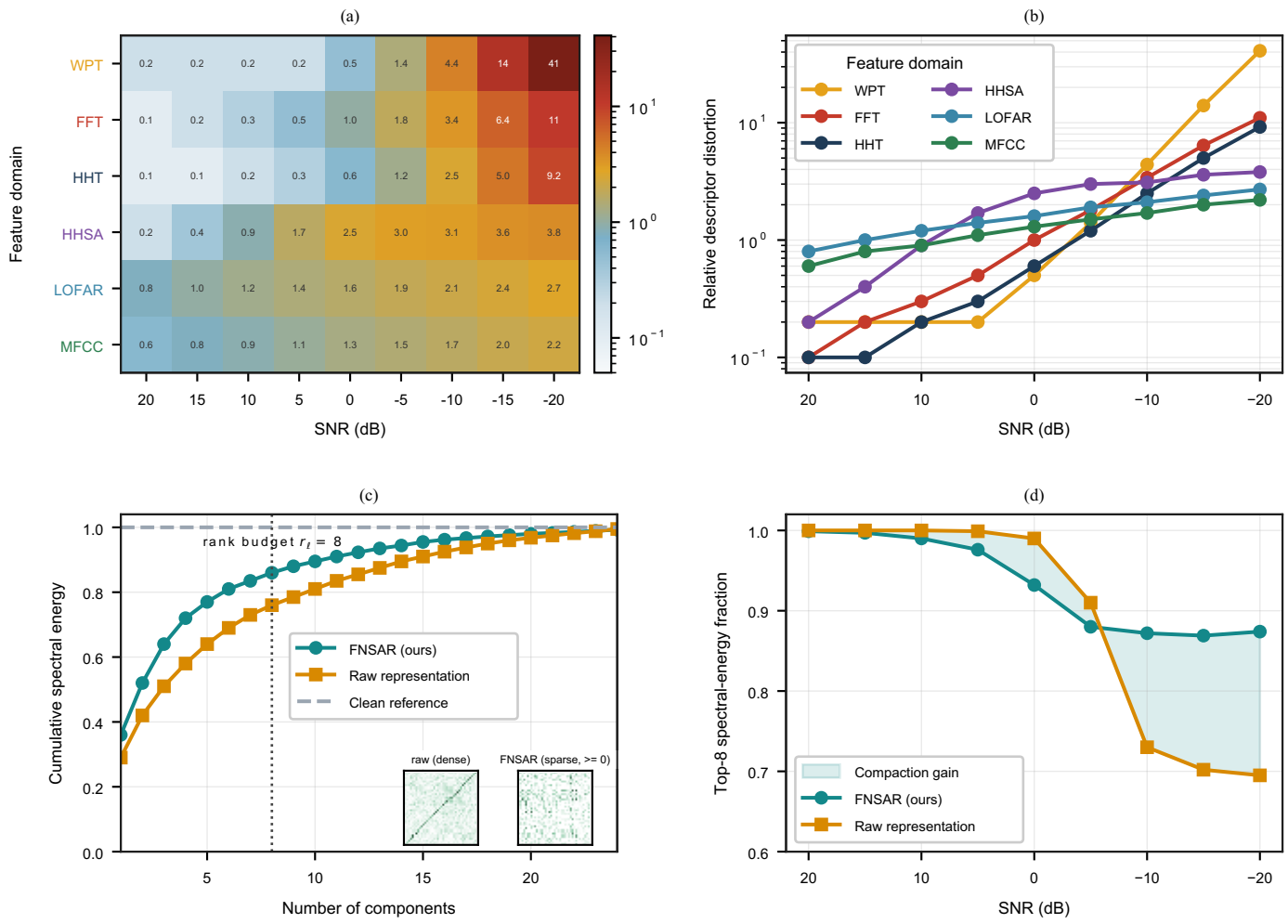
(a)
Feature domain
WPT
FFT
HHT
HHSA
LOFAR
MFCC
0.2 0.2 0.2 0.2 0.5 1.4 4.4 14 41
0.1 0.2 0.3 0.5 1.0 1.8 3.4 6.4 11
0.1 0.1 0.2 0.3 0.6 1.2 2.5 5.0 9.2
0.2 0.4 0.9 1.7 2.5 3.0 3.1 3.6 3.8
0.8 1.0 1.2 1.4 1.6 1.9 2.1 2.4 2.7
0.6 0.8 0.9 1.1 1.3 1.5 1.7 2.0 2.2
10^1
10^0
10^{-1}
20 15 10 5 0 -5 -10 -15 -20
SNR (dB)
(b)
Feature domain
WPT
FFT
HHT
HHSA
LOFAR
MFCC
Relative descriptor distortion
10^1
10^0
10^{-1}
20 10 0 −10 −20
SNR (dB)
(c)
rank budget $r_t = 8$
FNSAR (ours)
Raw representation
Clean reference
raw (dense)
FNSAR (sparse, >= 0)
Cumulative spectral energy
1.0 0.8 0.6 0.4 0.2 0.0
5 10 15 20
Number of components
(d)
Compaction gain
FNSAR (ours)
Raw representation
Top-8 spectral-energy fraction
1.0 0.9 0.8 0.7 0.6
20 10 0 −10 −20
SNR (dB)